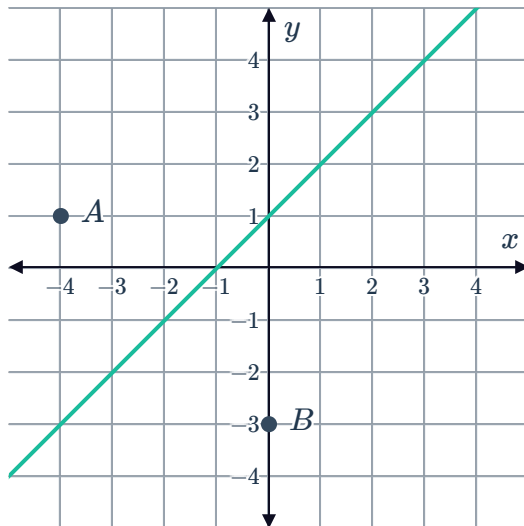


Construct Voronoi diagrams

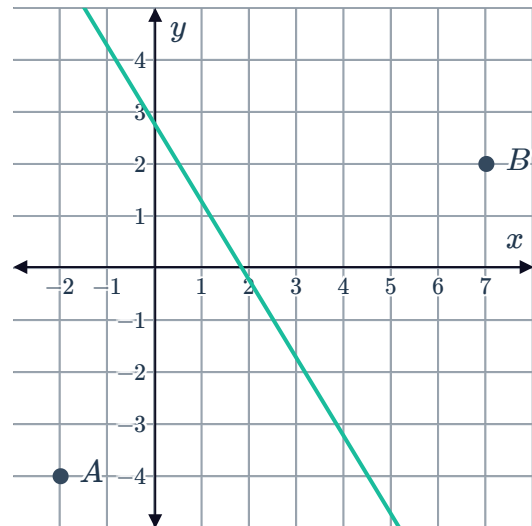


1

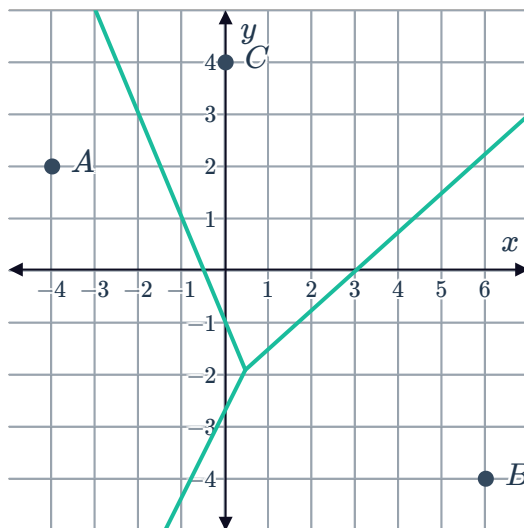
a



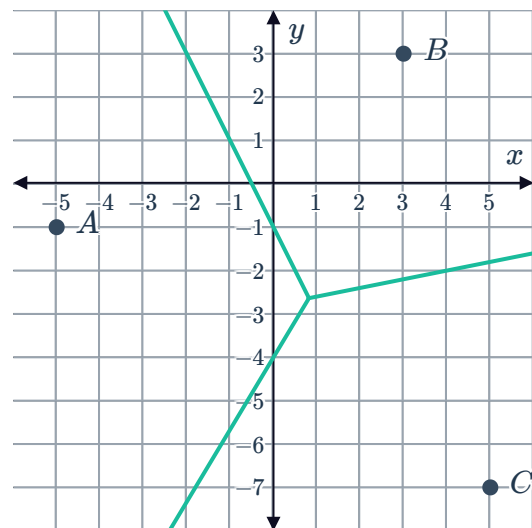
b



c

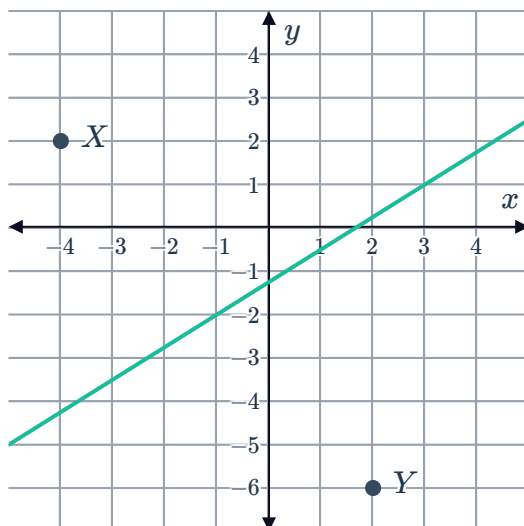


d



2

a



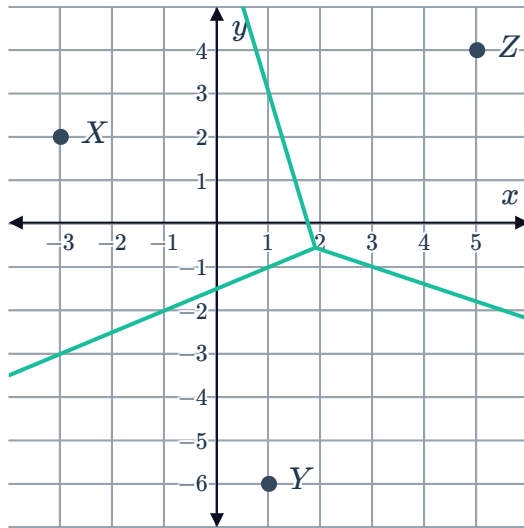
b Site X

c Example answers: $(-1, -2)$ $(2, 1)$



3

a

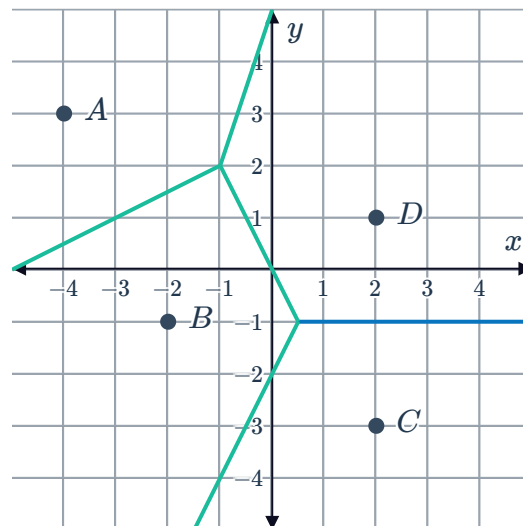


b Both site X and site Z .

4

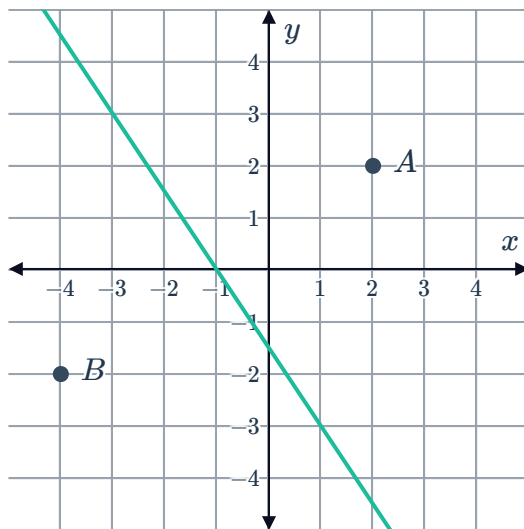
a Because sites C and D are in the same cell, so we need to add an edge that is equidistant between these two sites.

b

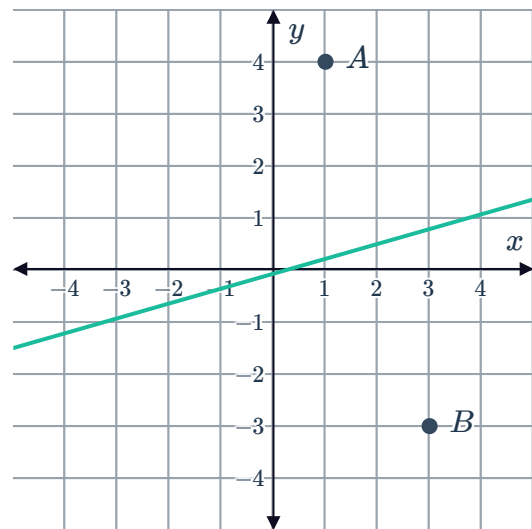


5

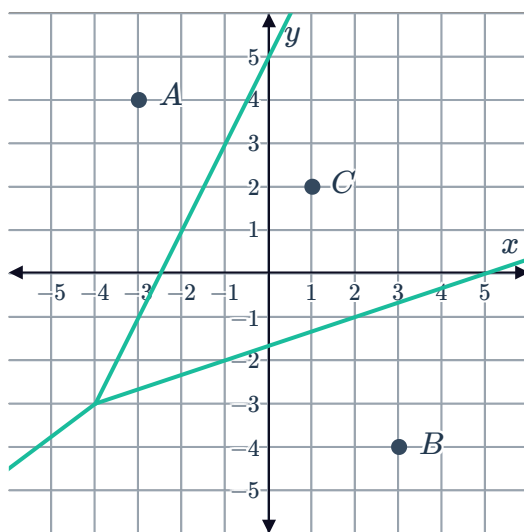
a



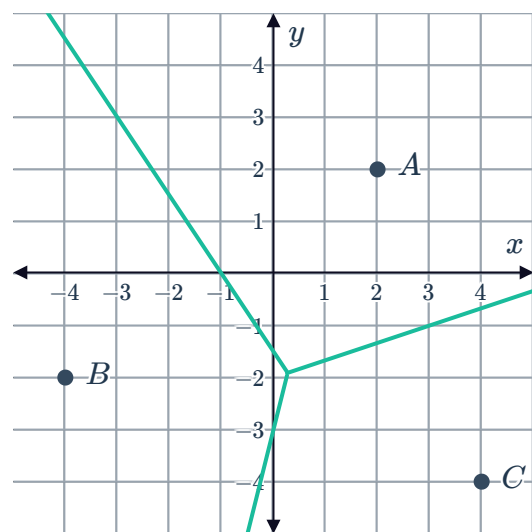
b



c



d

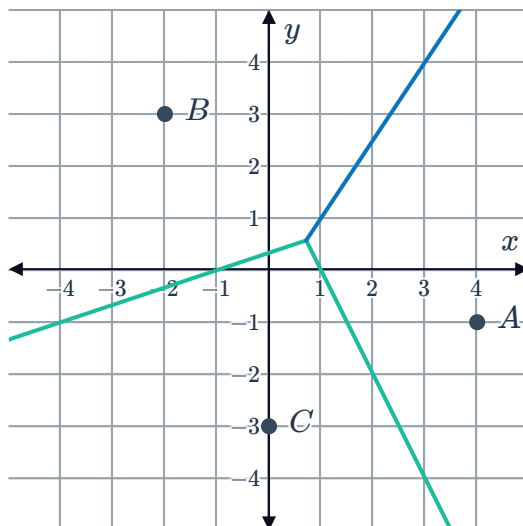


6

a

i $3x - 2y - 1 = 0$

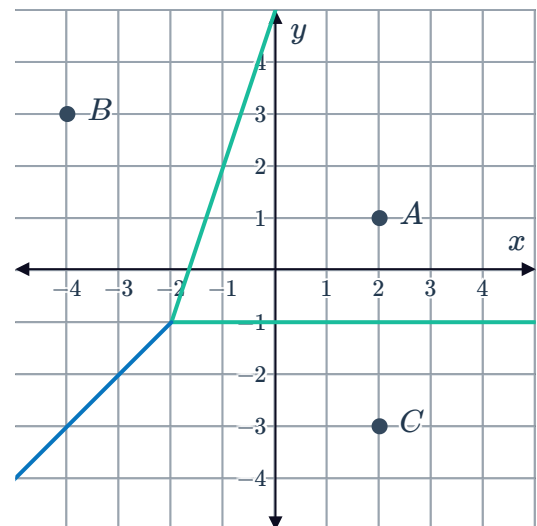
ii



b

i $y = x + 1$

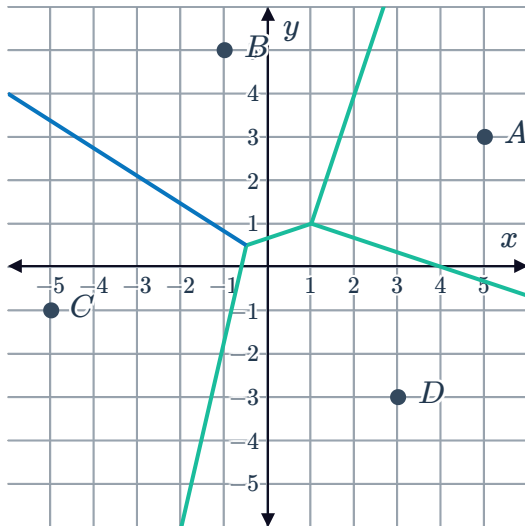
ii



c

i $7x + 11y - 2 = 0$

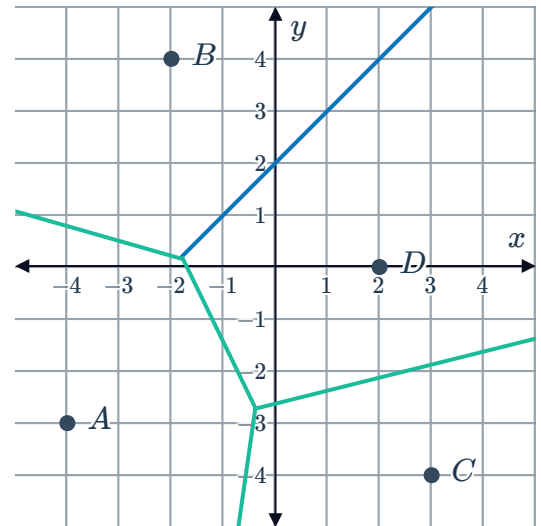
ii



d

i $y = x + 2$

ii



7

a Because there is a cell without a site.

b $(1, -1)$

8

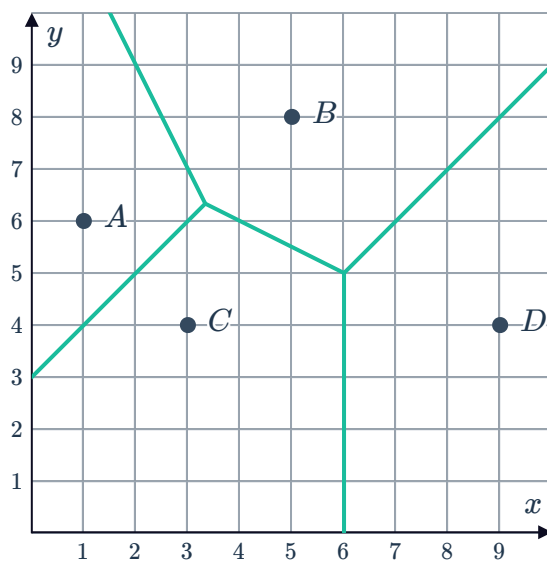
a $(4, 4)$

b $(-3, 2)$

Applications

9

a



b Bus stop C

10 $(-1, 2)$